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Next item →

1. Which one of the following is a correct definition of an exponential function of base b ?

1 / 1 point

- ☐ $f : \mathbb{R} \rightarrow \mathbb{R}^+$ and $f(x) = x^b$ where $b > 0$ and $b \neq 1$
- ☒ $f : \mathbb{R} \rightarrow \mathbb{R}^+$ and $f(x) = b^x$ where $b > 0$ and $b \neq 1$
- ☐ $f : \mathbb{R} \rightarrow \mathbb{R}^+$ and $f(x) = x^b$ where $b \geq 0$
- ☐ $f : \mathbb{R} \rightarrow \mathbb{R}^+$ and $f(x) = x^b$ where $b \leq 0$

✔ Nice work

This is definition of the exponential function with a base b .

2. Write down the value of $\log_2 128$.

1 / 1 point

7

✔ Nice work

$$\log_2 128 = \log_2 2^7 = 7 \log_2 2 = 7 \times 1 = 7$$

3. $\log_{100} x = 8$ is equivalent to:

1 / 1 point

- ☐ $8 = 100^x$
- ☐ $100 = x^8$
- ☐ $x = 8^{100}$
- ☒ $x = 100^8$

✔ Nice work

$$x = 100^x \implies \log_{100} x = \log_{100} 100^8 = 8 \log_{100} 100 = 8$$

4. Which one of the following curves represents the function $\left(\frac{1}{2}\right)^x$? Choose the correct label that applies.

1 / 1 point

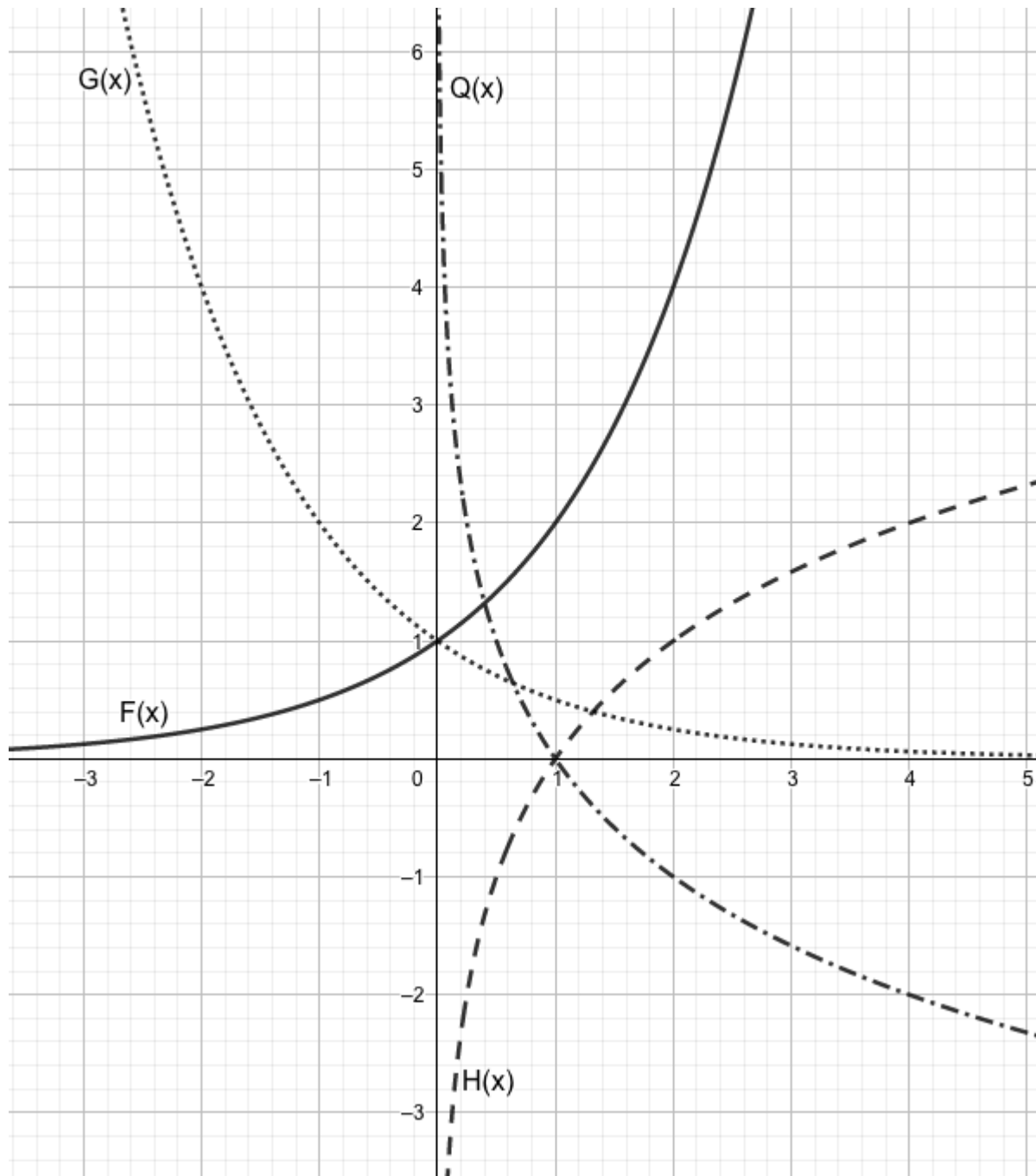


Image description:

The domain for the function Q is $(0, +\infty)$ and its corresponding curve is decreasing from $+\infty$ to $-\infty$ and crosses the x -axis on $x = 1$.

The domain for the function H is $(0, +\infty)$ and its corresponding curve is increasing from $-\infty$ to $+\infty$ and crosses the x -axis on $x = 1$.

The domain for the function F is $(-\infty, +\infty)$ and its corresponding curve is increasing from 0 to $+\infty$ and crosses the y -axis on $y = 1$.

The domain for the function G is $(-\infty, +\infty)$ and its corresponding curve is decreasing from $+\infty$ to 0 and crosses the y -axis on $y = 1$.

- ☐ $H(x)$
- ☐ $Q(x)$
- ☒ $G(x)$
- ☐ $F(x)$

✔ Nice work

The base is equal to $\frac{1}{2}$. Hence, the function is decreasing (exponential decay) traversing the y -axis at point $(0,1)$.